

3. Playfair Cipher

```
#include <stdio.h> #include <string.h> #include <ctype.h>

char matrix[5][5];

void createMatrix(char key[]) { int used[26] = {0}; int i, j, k = 0; char ch;

used['J' - 'A'] = 1;

/* Insert keyword */
for (i = 0; key[i] != '\0'; i++)
{
    ch = toupper(key[i]);

    if (ch == 'J')
        ch = 'I';

    if (ch >= 'A' && ch <= 'Z' && !used[ch - 'A'])
    {
        matrix[k / 5][k % 5] = ch;
        used[ch - 'A'] = 1;
        k++;
    }
}

/* Insert remaining alphabet */
for (ch = 'A'; ch <= 'Z'; ch++)
{
    if (!used[ch - 'A'])
    {
        matrix[k / 5][k % 5] = ch;
        used[ch - 'A'] = 1;
        k++;
    }
}

}
```

```

void findPosition(char ch, int *row, int *col) { int i, j;

if (ch == 'J')
    ch = 'I';

for (i = 0; i < 5; i++)
{
    for (j = 0; j < 5; j++)
    {
        if (matrix[i][j] == ch)
        {
            *row = i;
            *col = j;
            return;
        }
    }
}

}

int main() { char key[50], text[100], plain[200], cipher[200]; int i, len = 0, n = 0; int r1, c1, r2,
c2;

printf("Enter keyword: ");
scanf("%s", key);

printf("Enter plaintext: ");
scanf("%s", text);

createMatrix(key);

printf("\nPlayfair Matrix:\n");
for (i = 0; i < 5; i++)
{
    printf("%c %c %c %c %c\n",
        matrix[i][0], matrix[i][1], matrix[i][2],
        matrix[i][3], matrix[i][4]);
}

```

```

/* Prepare plaintext */
for (i = 0; text[i] != '\0'; i++)
{
    char ch = toupper(text[i]);

    if (ch == 'J')
        ch = 'I';

    if (ch >= 'A' && ch <= 'Z')
        plain[len++] = ch;
}

plain[len] = '\0';

/* Insert X between repeated letters */
for (i = 0; i < len; i++)
{
    cipher[n++] = plain[i];

    if (i + 1 < len && plain[i] == plain[i + 1])
        cipher[n++] = 'X';
}

if (n % 2 != 0)
    cipher[n++] = 'X';

cipher[n] = '\0';

/* Encrypt pairs */
for (i = 0; i < n; i += 2)
{
    findPosition(cipher[i], &r1, &c1);
    findPosition(cipher[i + 1], &r2, &c2);

    if (r1 == r2)
    {
        cipher[i] = matrix[r1][(c1 + 1) % 5];
        cipher[i + 1] = matrix[r2][(c2 + 1) % 5];
    }
}

```

```
    else if (c1 == c2)
    {
        cipher[i] = matrix[(r1 + 1) % 5][c1];
        cipher[i + 1] = matrix[(r2 + 1) % 5][c2];
    }
    else
    {
        cipher[i] = matrix[r1][c2];
        cipher[i + 1] = matrix[r2][c1];
    }
}

printf("\nCiphertext: %s\n", cipher);

return 0;

}
```



C:\Users\ebina\OneDrive\Des



```
Enter keyword: monarchy
Enter plaintext: instruments
```

Playfair Matrix:

```
M O N A R
C H Y B D
E F G I K
L P Q S T
U V W X Z
```

Ciphertext: GATLMZCLRQXA

```
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Process exited after 19.37 seconds with return value 0
Press any key to continue . . .
```